Supplemental Appendix

The Inequality Multiplier: Portfolio Allocation Frictions and Wealth Inequality

C. Christopher Hyland* Aditya Khemka†

The supplemental appendix provides institutional evidence and microfoundations (S1), a three-period illustration (S2), sensitivity to preferences, portfolio and borrowing assumptions (S3) and the complete impulse responses (S4).

S1. Institutional evidence and microfoundations

S2. A stylized three-period model

S3. Preference, portfolio and borrowing sensitivity

S4. Complete impulse responses

S1 Institutional Evidence and Microfoundations

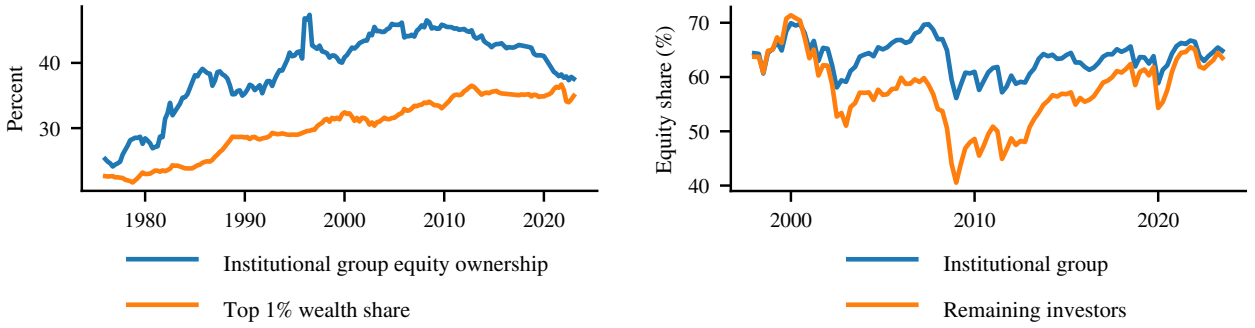
This section documents institutional ownership and portfolio composition and then discusses mechanisms that can limit portfolio substitution. The evidence is descriptive; the mechanisms motivate the allocation restriction used in the model.

S1.1 Institutional Ownership and Portfolio Composition

Figure 1 documents the ownership and portfolio patterns that motivate the model's allocation restriction. Panel (a) compares the top 1% wealth share with the fraction of corporate equity held by a selected institutional group comprising investment funds, insurance companies and retirement institutions. Panel (b) compares the equity share of this group's measured equity and bond portfolio with that of the remaining investors. The institutional group's equity share is higher and varies less over the plotted sample. This is consistent with a role for portfolio targets, although portfolio weights also reflect returns, funding and changes in investor composition.

*Christopher Hyland: Saïd Business School and Exeter College, University of Oxford and Oxford-Man Institute of Quantitative Finance (christopher.hyland.dphil@said.oxford.edu)

†Aditya Khemka (Corresponding Author): Saïd Business School and New College, University of Oxford (aditya.khemka.dphil@said.oxford.edu)



(a) Institutional ownership and wealth concentration

(b) Equity portfolio shares

Figure 1: Institutional equity ownership, wealth concentration and portfolio composition.

Note: Panel (a) shows the selected institutional group's share of corporate equity and the top 1% wealth share over 1976Q1–2023Q1. Panel (b) shows equity holdings divided by equity plus measured Treasury and corporate bond holdings for that group and the remaining investors over 1998Q1–2023Q3. These portfolio shares use the assets included in the calculation. The institutional group comprises property-casualty and life insurers, private pension funds, federal and state/local retirement funds, mutual funds, closed-end funds and ETFs. The same sectors enter equity and bonds, using sector totals without adding their components. Remaining investors are the all-sector totals less this group.

Source: Authors' calculations from the archived Financial Accounts workbook and quarterly wealth-share compilation. Portfolio shares cover equity, Treasury securities and corporate bonds; other assets are excluded from the denominator.

To relate portfolio composition to wealth concentration, Figure 2 plots changes in wealth shares against changes in the institutional group's bond-to-equity ratio, $B/(pQ)$. If a portfolio consists of equity and bonds with equity share θ, then

$$\frac{B}{pQ} = \frac{1-\theta}{\theta}, \qquad \epsilon_{\text{fixed share}} = 1 - \theta = \frac{B}{B + pQ},$$

where the second expression is the conditional quantity elasticity with a fixed portfolio share and inherited wealth revaluation, as derived in Appendix E.3. The bond-to-equity ratio is therefore informative about the portfolio weight entering that special case. Measuring demand elasticity also requires the response of allocation and funding to price.

The negative association is consistent with wealth concentration increasing when the institutional portfolio becomes more weighted toward equity. The descriptive fit is weak for the top 10%, with an R^2 of about 0.024, and stronger for the top 1%, at approximately 0.125. Equity appreciation itself can both lower $B/(pQ)$ and raise the wealth share of equity owners. These descriptive associations motivate studying portfolios and inequality jointly; the model examines how portfolio allocation restrictions affect wealth concentration.

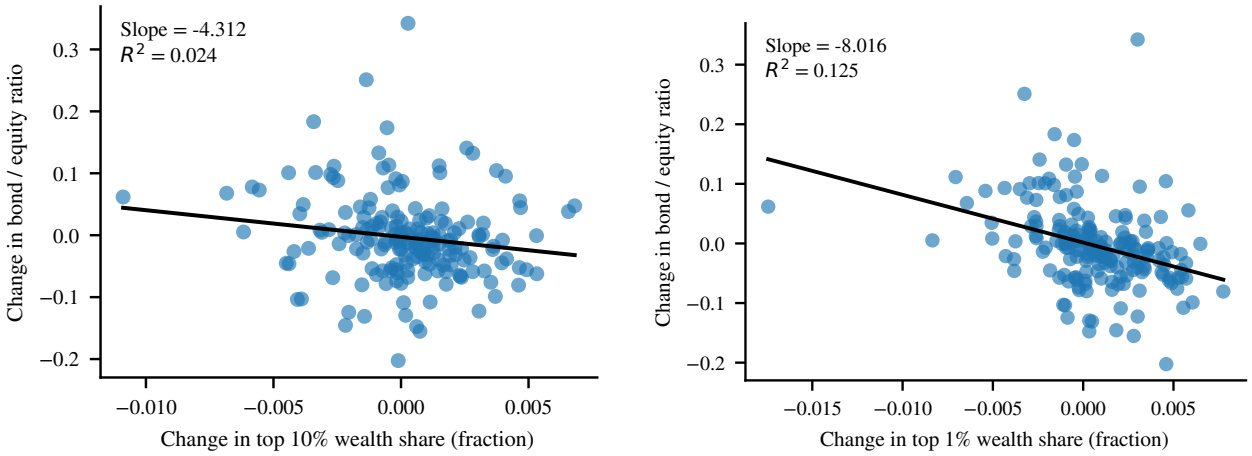


Figure 2: Changes in wealth concentration and institutional portfolio composition.

Note: Quarterly first differences over 1976Q2–2023Q1. The horizontal axes show changes in the top 10% and top 1% wealth shares, measured as fractions. The vertical axis shows the change in $B/(pQ)$ for the institutional group in Figure 1. Lines and R^2 summarize the descriptive linear fit.

Source: Authors' calculations from the Financial Accounts and the wealth share compilation used in Figure 1.

S1.2 Microfoundations for Market Inelasticity

Several features of asset markets can limit portfolio substitution: intermediary mandates and agency frictions, transaction costs, delays in moving capital, preferred-habitat demand and binding intermediary constraints. We review these motivations before relating them to the allocation restriction used in the model.

Mutual funds and ETFs are often subject to explicit or implicit investment mandates that limit portfolio discretion; Section **??** gives institutional examples. He and Xiong (2013) use an agency-based model to argue that tight investment mandates are justified for most fund managers, except those with exceptional talent, which explains why stringent investment mandates are observed in practice.

Other microfoundations include the transaction costs of moving wealth between liquid and illiquid assets in Kaplan et al. (2018), which reduce households' ability to adjust the shares of their portfolios invested in different asset classes. Slow-moving capital in Duffie (2010) (due to institutional impediments to trade such as search costs, time to raise capital or investor inattention) can generate sharp price reactions in the short to medium run. Preferred-habitat investors in Vayanos and Vila (2021) and Chen et al. (2012) create limits to arbitrage, which implies that asset markets are not fully elastic. More generally, behavioral frictions such as inattention may also help explain investors' limited ability to absorb large flows into and out of asset markets.

Haddad and Muir (2021) use time-varying risk aversion of financial intermediaries along with costly direct investment in asset markets by households to show that the elasticity of household asset demand to asset prices is decreasing in the risk aversion of financial intermediaries as well as the cost of direct investment, which generates inelastic demand curves. In a similar vein, He and Krishnamurthy (2018) review approaches under which institutional frictions such as equity capital constraints (He and Krishnamurthy, 2013), endogenous borrowing constraints (Brunnermeier and Sannikov, 2014), value-at-risk regulations (Adrian and Shin, 2010) and leverage regulations (Du et al., 2023) limit the responsiveness of asset demand, including at commercial banks, investment banks and hedge funds. A downward-sloping demand curve alone does not establish an elasticity below one.

For mutual funds and asset managers, related explanations emphasize managerial skill (Kacperczyk et al., 2016), limits to information acquisition (Gârleanu and Pedersen, 2018) and benchmarking considerations (Basak and Pavlova, 2013; Pavlova and Sikorskaya, 2023). These factors affect managers' incentives and ability to change their portfolios.

A natural question is why macro-arbitrageurs have not emerged to exploit these frictions and thereby offset the inelasticity they generate. Gabaix and Koijen (2021) document the limited size and influence of arbitrageurs, pointing to the small equity holdings of hedge funds relative to the market. This is evidence about one potential source of arbitrage capacity, rather than a measure of all investors able to trade against price pressure. They also note that hedge funds seem to respond to crisis episodes by selling assets rather than buying, that the transfer of equity risk across different types of investors is minimal and that households' equity share has shown little variation over time. Finally, Haddad et al. (2025) highlight that the strategic response of investors to others' reduced demand elasticity is limited, which further contributes to market inelasticity. These findings suggest that the forces that might offset market inelasticity are limited.

These mechanisms motivate studying asset markets in which investors have limited scope to change their portfolios. Our constrained fund captures this restriction through a cost of deviating from a target allocation. The flexible benchmark removes that target penalty while retaining costs of equity and bond exposures, as described in Appendix **??**. Those costs depend on position levels and support the reference equity premium; they do not impose a target share or a delay in changing it. Appendix E.3 derives how the two specifications differ in conditional demand elasticity.

The resulting borrowing and wealth responses follow from the particular allocation and household portfolio restrictions in our model. Other sources of limited substitution can affect financing, risk sharing and price adjustment differently. Appendix D examines a

trader with finite capacity to absorb valuation differences, using the flexible benchmark as its reference.

S2 A Stylized Three-Period Model

This section presents the simplest version of the mechanism in Section 3: two household types, two assets and a mandate-constrained intermediary, over three dates. It strips out housing, the middle class, non-homothetic preferences and the borrowing frictions of the quantitative model, retaining only the portfolio mandate that ties the equity price to the outstanding stock of debt. The resulting pricing identity, $p_0 = p_{-1} + \frac{\theta}{1-\theta}\frac{B_0^P}{\bar{Q}}$, is the exact three-date analogue of the quantitative mandate identity $p_t\bar{Q} = \frac{\theta_t}{1-\theta_t}B_t^{MF}$.

The wealthy household (R) saves only through the intermediary, which allocates its wealth between equity and debt; the non-wealthy household (P) receives labor income and can only borrow, because it is institutionally excluded from equity. The intermediary (a mutual fund) is required to hold a fixed proportion of equity and bonds, which is the sole friction. Figure 3 illustrates the setup.

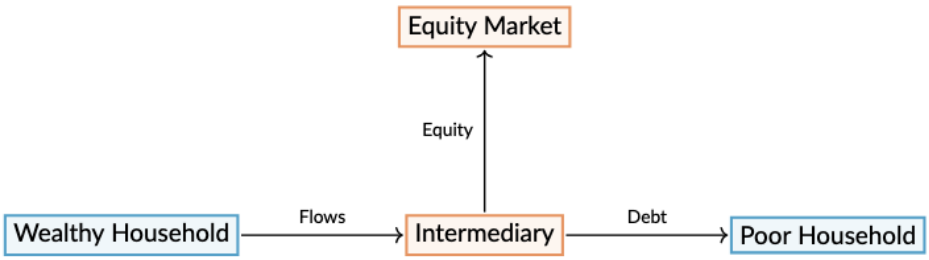


Figure 3: Environment

Timing. At $t = -1$, the intermediary allocates initial wealth between equity and bonds without a mandate. The wealthy own this portfolio, and the non-wealthy owe the debt in its bond position. At $t = 0$, both households choose consumption and saving for dates 0 and 1. The wealthy invest F_0, the non-wealthy take on new debt B_0^P, and the intermediary must maintain its inherited equity share. With equity supply fixed, the price adjusts to clear the market. At $t = 1$, the equity dividend is paid, the portfolio unwinds and debts are

repaid. Figure 4 summarizes the timing.

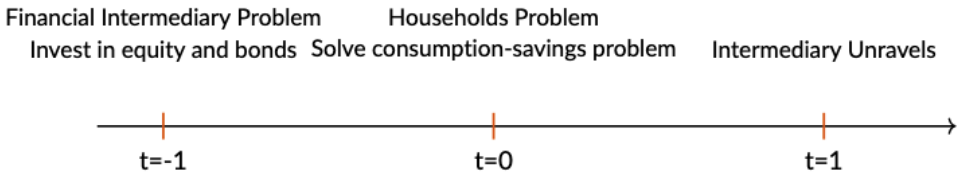


Figure 4: Timeline

S2.1 Financial Intermediary

At $t = -1$, given initial wealth W_{-1}, dividend D_1 and the two-period annualized rate $r_{f,-1}$, the intermediary chooses its equity share θ to maximize returns on final-period wealth:

$$\max_{\theta} \frac{1}{(1 + r_{f,-1})^2}\left[\theta \frac{D_1}{p_{-1}} + (1 - \theta)(1 + r_{f,-1})^2\right]. \tag{1}$$

The first-order condition yields the fundamental price at $t = -1$:

$$p_{-1} = \frac{D_1}{(1 + r_{f,-1})^2}. \tag{2}$$

We use a single intermediary with equity share θ. For heterogeneous intermediaries with assets W_i, the consolidated share is $\sum_i W_i\theta_i / \sum_i W_i$; equity-holding weights instead enter the aggregation of conditional demand elasticities (Gabaix and Koijen, 2021). Holding the representative share fixed isolates the portfolio mechanism. At $t = -1$, the accounting identities are

$$\theta W_{-1} = p_{-1}Q_{-1}, \qquad (1 - \theta)W_{-1} = B_{-1}. \tag{3}$$

At $t = 0$, the intermediary receives an inflow F_0 and the price adjusts to p_0, but it cannot readjust θ. Its holdings satisfy

$$\theta\Big(p_0Q_{-1} + B_{-1} + F_0\Big) = p_0Q_0(F_0), \qquad (1 - \theta)\Big(p_0Q_{-1} + B_{-1} + F_0\Big) = B_0(F_0), \tag{4}$$

where $Q_0(F_0)$ and $B_0(F_0)$ are the new equity and bond quantities. Intermediary wealth is

$$W_0 = p_0 Q_{-1} + B_{-1} + F_0 = p_0 Q_0 + B_0, \tag{5}$$

so flows are absorbed as new equity plus new bonds:

$$F_0 = p_0(Q_0 - Q_{-1}) + (B_0 - B_{-1}). \tag{6}$$

The $t = 0$ portfolio rigidity captures institutional mandates or costly rebalancing; potential microfoundations are reviewed in Supplemental Appendix S1.2. Appendix D examines an arbitrageur extension of the quantitative model.

S2.2 Households

Non-Wealthy Household. The non-wealthy household chooses $\{c_0^P, c_1^P, B_0^P\}$ to solve max $u[c_0^P] + \beta u[c_1^P]$, given endowments $\{e_0^P, e_1^P\}$ and legacy debt $(1 + r_{f,-1})^2 B_{-1}$,[1] subject to

$$c_0^P \le e_0^P + B_0^P, \qquad c_1^P + (1 + r_{f,-1})^2 B_{-1} + (1 + r_{f,0}) B_0^P \le e_1^P. \tag{7}$$

Its Euler equation prices the risk-free bond.

Wealthy Household. The wealthy household chooses $\{c_0^R, c_1^R, F_0\}$ to solve max $u[c_0^R] + \beta u[c_1^R]$, given $\{e_0^R, e_1^R\}$ and end-of-period income $D_1 Q_0 + (1 + \tilde{r}_{f,0}) B_0$, subject to

$$c_0^R + F_0 \le e_0^R, \qquad c_1^R \le e_1^R + D_1 Q_0(F_0) + (1 + \tilde{r}_{f,0}) B_0(F_0), \tag{8}$$

where $1 + \tilde{r}_{f,0} = (1 + r_{f,-1})^2 B_{-1}/B_0 + (1 + r_{f,0}) B_0^P / B_0$. Its Euler equation prices returns on the intermediary, not equity directly. Equity clears at fixed supply, $Q_{-1} = Q_0 = \bar{Q}$.

S2.3 Equilibrium

Definition S2.1. *Given initial portfolios and endowments, an equilibrium consists of household consumption and saving, intermediary holdings and allocation, and asset prices such that households*

[1] Without loss of generality, this is two-period debt with no intermediate coupon.

and the intermediary optimize and markets clear:

$$Q_{-1} = Q_0 = \bar{Q}, \qquad B_0 = B_0^P + B_{-1},$$
$$c_0^R + c_0^P = e_0^R + e_0^P, \qquad c_1^R + c_1^P = e_1^R + e_1^P + D_1\bar{Q}.$$

S2.4 Equilibrium Pricing

Imposing equity-market clearing yields the central pricing result. Because flows from the wealthy and borrowing by the non-wealthy are two sides of the same transaction, the result is stated directly in terms of the bond position.

Proposition S2.1. *The price of equity at $t = 0$ is*

$$p_0 = p_{-1} + \frac{\theta}{1-\theta}\frac{B_0^P}{\bar{Q}}, \tag{9}$$

where p_{-1} is the initial discounted dividend value and $\frac{\theta}{1-\theta}\frac{B_0^P}{\bar{Q}}$ is the adjustment associated with new lending. Income fundamentals can drive this lending and hence both terms in the price decomposition.

Proof. Equity-market clearing requires $(Q_0 - Q_{-1})/Q_{-1} = 0$. Substituting $Q_0 = \frac{\theta}{p_0}(p_0 Q_{-1} + B_{-1} + F_0)$ from (4),

$$\theta + \frac{\theta B_{-1}}{p_0 Q_{-1}} + \frac{\theta F_0}{p_0 Q_{-1}} = 1.$$

Using $B_{-1} = (1-\theta)W_{-1}$ and $Q_{-1} = \theta W_{-1}/p_{-1}$ from (3),

$$\theta p_0 + (1-\theta)p_{-1} + \frac{\theta F_0}{Q_{-1}} = p_0 \implies p_0 = p_{-1} + \frac{\theta}{1-\theta}\frac{F_0}{Q_{-1}}.$$

From (6) with $Q_0 = Q_{-1} = \bar{Q}$, $F_0 = B_0 - B_{-1}$; since the bond leg grows only through new lending to the non-wealthy, $B_0 - B_{-1} = B_0^P$, so $F_0 = B_0^P$. Substituting and imposing $Q_{-1} = \bar{Q}$ gives (9). □

For every additional dollar of borrowing absorbed by the intermediary, equity market value must rise by $\frac{\theta}{1-\theta} = 3.55$ at the calibrated $\theta = 0.78$; we call $\frac{\theta}{1-\theta}$ the *price impact* of borrowing. There is a single mechanism: saving by the wealthy is intermediated as new lending to the non-wealthy, so $F_0 = B_0^P$; this dollar cannot buy more equity (supply is fixed at $\bar{Q}$), so it is absorbed as a bond, but the mandate forces the equity leg $p_0\bar{Q}$ to rise in step. With fixed supply, the equity price is pinned to the outstanding bond position rather than to fundamentals alone.

With log utility, borrowing depends on endowments, inherited debt and the safe rate:

$$F_0 = B_0^P = \frac{e_1^P - \beta(1 + r_{f,0})e_0^P - (1 + r_{f,-1})^2 B_{-1}}{(1 + \beta)(1 + r_{f,0})}, \tag{10}$$

so a fall in the non-wealthy income share need not reduce borrowing if their future income remains sufficient to cover repayment and desired consumption.

S2.5 Price Decomposition and Downward-Sloping Demand

The borrowing solution makes the role of endowments explicit. Write $A = (1 + r_{f,-1})^2$ for the gross return on legacy debt and $R = 1 + r_{f,0}$ for the return on new lending.

Proposition S2.2. *Under log utility, the equity price satisfies*

$$p_0 = p_{-1}\left(1 - \frac{A}{(1 + \beta)R}\right) + \frac{1}{1 + \beta}\frac{\theta}{1 - \theta}\frac{1}{\bar{Q}}\left(\frac{e_1^P}{R} - \beta e_0^P\right), \qquad p_{-1} = \frac{D_1}{A}. \tag{11}$$

Proof. The non-wealthy Euler equation gives $c_1^P = \beta R c_0^P$. Substituting into its intertemporal budget yields

$$B_0^P = c_0^P - e_0^P = \frac{e_1^P - AB_{-1} - \beta R e_0^P}{(1 + \beta)R}.$$

Insert this expression into (9) and use $B_{-1} = (1 - \theta)p_{-1}\bar{Q}/\theta$ to obtain the result. □

The formula retains the returns on both legacy and new debt; no restriction equating them is needed. New borrowing is positive when $e_1^P > AB_{-1} + \beta R e_0^P$. Higher future income supports borrowing, while current income and inherited repayment obligations reduce it, conditional on the safe rate. With fixed allocation, the associated lending changes equity value. Household endowments can therefore affect prices through portfolio balance as well as through discount rates. A higher θ increases the price impact of a given dollar of lending. Figure 5 contrasts this conditional demand schedule with the flat schedule of the flexible-allocation example below.

S2.6 Elasticity of Equity Demand

With net funding fixed and inherited equity allowed to revalue, the positive quantity elasticity is pinned down by θ. It is below one for an interior equity share.

Proposition S2.3. *The conditional quantity elasticity of equity demand is* $\xi = 1 - \theta$.

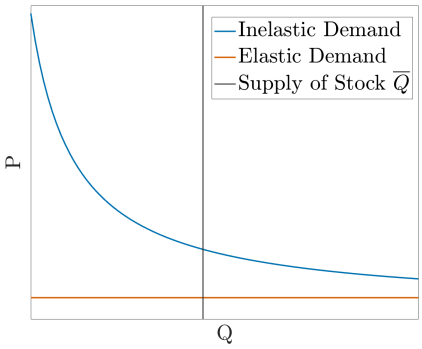


Figure 5: Inelastic demand curve ($\xi = 0.2$) versus elastic demand curve.

Proof. From (4) and (5), $Q_0 = \theta W_0/p_0$. Taking logs and first-differencing, $\Delta \ln Q = \Delta \ln \theta + \Delta \ln W - \Delta \ln p$. A first-order approximation gives $\Delta \ln W \approx \theta \Delta \ln p + F_0/W_{-1}$; with $\Delta \ln \theta = 0$,

$$\Delta \ln Q = (\theta - 1)\Delta \ln p + \frac{F_0}{W_{-1}} \implies \xi = -\frac{\partial \Delta \ln Q}{\partial \Delta \ln p} = 1 - \theta. \qquad \square$$

As θ rises, demand becomes less elastic and the price impact $\theta/(1-\theta)$ grows. The limit $\theta \downarrow 0$ is not an interior allocation with positive equity value at fixed fund wealth. Nor is it the flexible benchmark: the fixed-share elasticity tends to one rather than infinity. Appendix S2.10 instead allows allocation to adjust at $t = 0$.

S2.7 Wealth Inequality

Definition S2.2. *For $t \in \{-1, 0\}$, wealth is $W_t^R \equiv p_t Q_t + B_t$ (wealthy) and $W_t^P \equiv O^P - B_t$ with $O^P > B_t$ (non-wealthy), where O^P denotes non-marketable assets (e.g., a primary residence) ensuring a non-negative net position. Wealth shares are $\eta_t^R \equiv W_t^R/(W_t^R + W_t^P)$ and $\eta_t^P \equiv 1 - \eta_t^R$.*

We consider $t \in \{-1, 0\}$ because the wealth evolution from $t = -1$ to $t = 0$ is what inelastic markets shape (at $t = 1$, wealth trivially equals income).

Proposition S2.4. *Holding O^P and equity supply fixed, the wealth share of the wealthy increases with the equity price along the fixed-allocation balance sheet:*

$$\frac{d\eta_0^R}{dp_0} = \frac{\bar{Q} O^P}{\theta(p_0 \bar{Q} + O^P)^2} > 0. \tag{12}$$

Proof. Fixed allocation implies $B_0 = (1-\theta)p_0\bar{Q}/\theta$. Thus

$$\eta_0^R = \frac{p_0\bar{Q}/\theta}{p_0\bar{Q} + O^P},$$

whose derivative gives the result. □

The derivative includes the additional lending required by the mandate. Holding bonds fixed would instead give $\bar{Q}(O^P - B_0)/(p_0\bar{Q} + O^P)^2$, the pure valuation effect. This is the same distinction underlying the conditional multipliers in Section **??**. The outside asset O^P enters measured wealth but provides no income in this example; it is held fixed across the comparisons.

S2.8 Comparative Statics on Income

Income affects prices through desired saving and borrowing. At a given safe rate, the borrowing expression above falls with current non-wealthy income and rises with future income. Wealthy households have the corresponding incentive to save more when current resources rise or expected future resources fall. In equilibrium, returns also adjust, so the net price and wealth-share effects depend on the timing and source of income changes.

Definition S2.3. *For $t \in \{0,1\}$, income is $I_t^R = e_t^R + \mathbb{I}_{t=1}D_1\bar{Q}$ and $I_t^P = e_t^P$, with shares $\mathcal{I}_t^i = I_t^i/(I_t^R + I_t^P)$.*

An illustration shows why income and wealth concentration need not move together. Set $D_1 = 2$, $\theta = 0.8$, $\beta = 0.99$, $W_{-1} = 5$, $\bar{Q} = 10$ and $O^P = 4$. Initial endowments are $e_0^P = 2$ and $e_0^R = 8$, giving $\mathcal{I}_0^R = 0.8$. The initial portfolio implies $p_{-1} = 0.4$, $B_{-1} = 1$ and $A = D_1/p_{-1} = 5$. With $e_1^P = 8$ and $e_1^R = 12$, the future wealthy income share is also 0.8. Solving the household Euler equations and market clearing conditions gives $B_0^P \simeq 0.063$ and $p_0 \simeq 0.425$. The wealthy household's wealth share rises from 0.625 to 0.644, or about 1.9 percentage points.

Now lower only future wealthy labor income to $e_1^R = 4$. Including dividends, the future wealthy income share falls to 0.75. Desired saving rises, new borrowing increases to approximately 0.314, and the equity price rises to 0.526. The wealthy household's wealth share reaches 0.710, about 8.5 percentage points above its initial value. The fall in future wealthy income induces saving today, and the mandate links the resulting lending to higher equity value. Non-wealthy income still rises from 2 to 8 across dates, which supports repayment. This experiment changes future aggregate resources; it is not a contemporaneous redistribution holding total income fixed. The quantitative income experiments in Appendix **??** distinguish these sources explicitly.

S2.9 Exogenous Increase in Equity Supply

The accounting mechanism extends to an exogenous supply change, although its quantitative strength can change. Let supply change by δ_q, so $Q_0 = Q_{-1}(1 + \delta_q)$.

Proposition S2.5. *The price of equity at $t = 0$ is*

$$p_0 = \left(\frac{1-\theta}{1-\theta+\delta_q}\right)\left[p_{-1} + \frac{\theta}{1-\theta}\frac{F_0}{Q_{-1}}\right]. \tag{13}$$

Proof. Clearing now requires $(Q_0 - Q_{-1})/Q_{-1} = \delta_q$. Repeating the proof of Proposition S2.1 with this right-hand side gives

$$(1-\theta)p_{-1} + \frac{\theta F_0}{Q_{-1}} = p_0(1-\theta+\delta_q),$$

which rearranges to (13). □

Since $\partial p_0/\partial \delta_q < 0$, the price multiplier $\frac{\theta}{1-\theta+\delta_q}$ is dampened for $\delta_q > 0$, but prices remain flow-sensitive unless $\theta = 0$. The percentage price change is $\frac{p_0-p_{-1}}{p_{-1}} = \frac{\theta F_0/(p_{-1}Q_{-1})-\delta_q}{1-\theta+\delta_q}$, so $\Delta p > 0 \Leftarrow \delta_q < \theta F_0/(p_{-1}Q_{-1})$. Writing flows as a share f of the intermediary's assets under management and using $p_{-1}Q_{-1} = \theta W_{-1}$, this is simply $\delta_q < f$. For the illustrative quarterly inputs $f = 2\%$ of initial fund assets and $\delta_q = 0.2\%$ of shares, issuance attenuates rather than eliminates the price effect. At $\theta = 0.78$, the response is approximately 8.1%, against 9.1% with fixed supply. Appendix **??** discusses the scope of this calculation. Finally, with variable supply, the tight B_0^P–F_0 link loosens, since part of each flow finances new shares:

Corollary S2.1. *When supply changes by δ_q, the price–debt relationship is*

$$p_0 = \left(\frac{1}{1+\delta_q}\right)\left[p_{-1} + \frac{\theta}{1-\theta}\frac{B_0^P}{Q_{-1}}\right]. \tag{14}$$

Proof. Substitute $F_0 = p_0(Q_0 - Q_{-1}) + (B_0 - B_{-1})$ into (13) and solve for p_0. □

As δ_q rises, the equity price impact of higher indebtedness falls.

S2.10 The Lucas Model as a Special Case

If instead the intermediary re-optimizes θ at $t = 0$, it solves $\max_\theta \frac{1}{1+r_{f,0}}[\theta\frac{D_1}{p_0} + (1-\theta)(1+r_{f,0})]$. The first-order condition with market clearing (no trade) gives $p_0 = D_1/(1 + r_{f,0})$. This deterministic, risk-neutral example assumes an interior portfolio with both assets held. Allowing rebalancing restores discounted-payoff pricing in this special case; it does not

reproduce the full stochastic Lucas economy or the finite exposure costs of the quantitative flexible benchmark.

S3 Sensitivity to Preferences, Portfolio Rigidity and Borrowing Constraints

This section assesses sensitivity to the portfolio mandate, the assignment of the household borrowing constraint and consumption curvature. The mandate and curvature exercises follow the original reference calibration procedure, while the borrower switch preserves preferences and the initial allocation.

S3.1 Sensitivity to the Portfolio Mandate

We assess how the simulated 1989–2023 wealth share transition depends on the target equity share θ^* and the cost of deviating from the target, χ^{MF}. For each value, the exercise repeats the reference calibration and the perfect-foresight transition using the same observed driver paths. The reference calibration continues to target the 1989 wealth shares and the household debt moment, so the jointly fitted parameters can adjust. The exercise therefore evaluates sensitivity after recalibration.

For $\chi^{MF} \in \{20, 171, 500\}$, the top share increases by approximately 5.34, 5.42 and 5.42 percentage points at the end of the transition. The corresponding bottom share changes are 1.17, 1.02 and 1.01 percentage points. The paths are close over this range, with the larger difference between 20 and 171. These results suggest that the historical comparison is not very sensitive to the precise target penalty once it is sufficiently large; allocation still responds somewhat at each finite value.

The target equity share has a larger effect on the distribution of wealth outside the top decile. Figure 6 compares $\theta^* \in \{0.70, 0.78, 0.86\}$. A lower target gives the fund a larger bond weight and changes the balance-sheet relation linking borrowing to equity value. In the simulated paths, the bottom share trough is approximately -4.9 percentage points at 0.70 and -2.8 at 0.86. Its terminal increase is approximately 2.31 versus 0.57 percentage points. The top share changes by 5.38 versus 5.91 percentage points. Thus the top share result is relatively stable across these targets, while the bottom share path is more sensitive. These comparisons reflect the combined adjustment of portfolios, borrowing and the recalibrated reference.

S3.2 Which Household Faces the Collateral Constraint?

The main model assigns a binding housing collateral constraint to the middle class and a convex borrowing cost to the non-wealthy. We examine whether the results depend on this assignment by reversing the borrowing arrangements. The non-wealthy now borrow

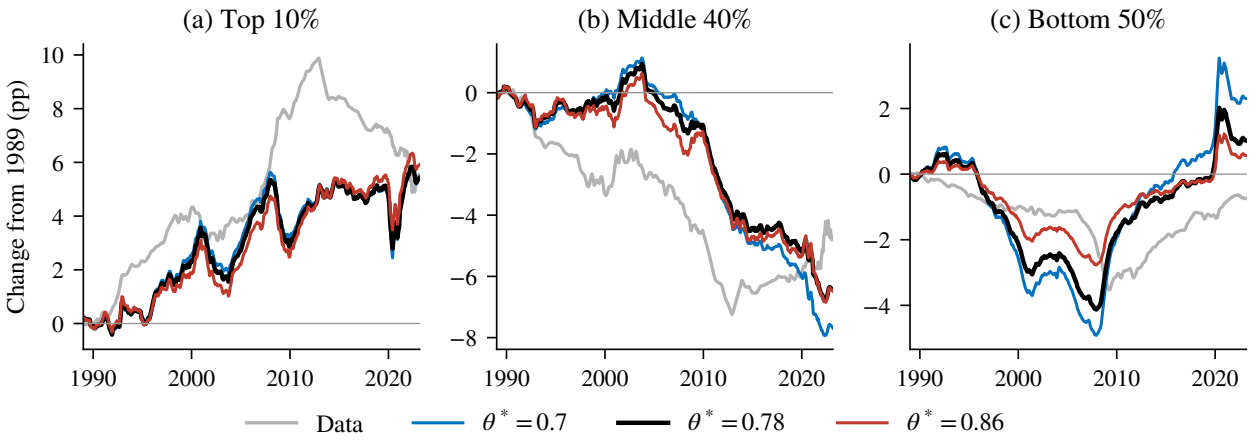


Figure 6: Sensitivity of the simulated wealth-share transition to the fund's target equity share θ^*.

Note: Perfect-foresight simulation of the top-10%, middle-40% and bottom-50% wealth shares (percentage-point change from 1989) for $\theta^* = 0.70$ (below calibration), 0.78 (calibrated, bold black) and 0.86 (above), against the data (gray). Each path repeats the reference calibration at the given θ^*, targeting the same initial wealth shares and debt moment, and then solves the transition under the same observed driver paths. Jointly fitted parameters can change across reference calibrations.

at a fixed loan-to-value ratio, while the middle class adjusts debt subject to a convex holding cost. Preferences and the initial allocation, including asset prices, wealth shares and consumption, are unchanged. We recover the new debt and housing costs from the reference Euler equations.[2]

Figure 7 compares wealth shares under unanticipated historical innovations. We use the same six paths for group labor incomes, equity dividends, housing rents and public debt in both specifications. The middle-class mortgage-limit driver is set to zero in both, since it has no effect when the middle-class collateral constraint is slack. The non-wealthy limit is constant. The figure displays the period through 2019.[3]

The rise in the top wealth share is similar, while its counterpart at the bottom changes substantially. In the constrained economy, the top share increases by about 6.8 percentage

[2]The non-wealthy loan-to-value ratio is their reference debt divided by housing value, $\tau^P = 0.8444$. Their debt-holding cost is removed. The middle-class debt cost takes the form $\Gamma_B^M(B) = \phi_B^M B^2/2$, with $\phi_B^M = (1 - \beta^M \bar{R})/\bar{B}^M \simeq 0.0126$. The housing cost coefficients become $\phi_H^M \simeq 0.2063$ and $\phi_H^P \simeq 0.2403$, satisfying the housing Euler equations at the same holdings and prices. Both fund economies admit determinate local solutions.

[3]In the full 1989–2023 linear simulations, the approximation to the non-wealthy collateral multiplier becomes negative in one month under unanticipated shocks in the switched constrained economy, and in two months in the switched flexible benchmark. The displayed period precedes these COVID-era violations of the maintained binding regime. Extending the comparison through those episodes would require allowing the constraint to become slack.

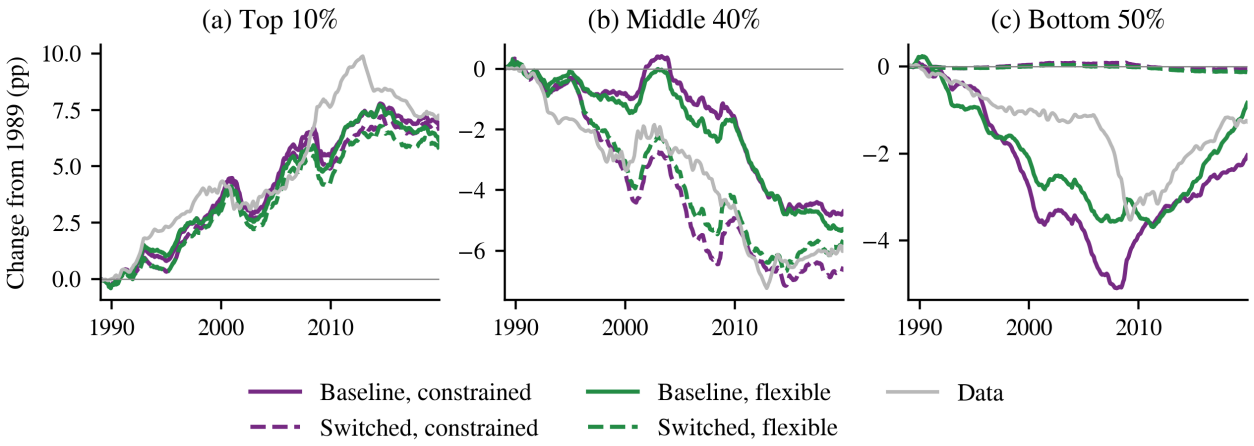


Figure 7: Wealth shares when the identity of the constrained borrower changes.

Note: Changes in wealth shares from 1989, in percentage points, under unanticipated innovations over 1989–2019. Solid lines use the main borrowing arrangements; dashed lines switch the binding collateral constraint to the non-wealthy and the convex borrowing cost to the middle class. Purple denotes the constrained economy and green the flexible benchmark; gray is the wealth-share data used in the main historical exercise. Both specifications use identical histories for the six common drivers, with the middle-class mortgage-limit driver held fixed.

points through 2019 under the main borrowing arrangements and 6.6 after the switch. The bottom share falls by about 2.0 percentage points in the main model but only 0.06 after the switch. Aggregate borrowing changes little, but the middle class absorbs much more of the credit expansion. Allowing non-wealthy borrowing to adjust independently of housing is therefore consequential for the model's account of declining net worth at the bottom.

The reason follows from the non-wealthy household's balance sheet. With all its debt at a binding ratio $\tau^P < 1$, its net worth equals $(1 - \tau^P)X_t^{H,P}$. Housing and debt move together, leaving a positive housing equity cushion and preventing debt from accumulating independently of collateral. This restriction substantially compresses the bottom wealth-share response in this calibration. The main model permits income deterioration and additional borrowing to reduce net worth beyond the movement in housing values.

We also repeat the demand experiments in Section **??**, keeping their innovation sizes and persistence fixed. Under the switch, the constrained economy's cumulative response to equity demand increases by about 2% and remains larger and longer-lasting than in the flexible benchmark. The constrained-to-flexible cumulative ratio rises from about 2.4 to 3.1. Broad fund demand continues to produce close cumulative responses, with the ratio near one. Thus the distinction between portfolio reallocation and demand for a larger financial portfolio survives the borrower switch. Responses to wealthy labor income and dividends are also close to those in the main specification. Redistribution is more sensitive:

its cumulative top-share response is about 14% smaller in the constrained economy than in the flexible benchmark in the main model, but about 7% larger after the switch. The response to the non-wealthy household's income loss depends on its borrowing margin.

The baseline borrowing arrangements therefore matter for the incidence of inequality, even though the principal comparison between the constrained economy and the flexible benchmark survives.

S3.3 Consumption Curvature and Historical Dynamics

The consumption elasticity of intertemporal substitution (EIS) is $1/\sigma_i$. The baseline assigns an EIS of approximately 0.24 to wealthy households and 0.70 to middle-class and non-wealthy households. We examine whether the historical comparison depends on these levels and their ordering.[4]

The first alternative raises $\sigma_M = \sigma_P$ to 5, keeping $\sigma_R = 4.1$. The borrowers' EIS is then 0.20, below that of wealthy households. This is an illustrative reversal of the ordering. The second sets $\sigma_R = 1/1.2 \simeq 0.833$ and leaves $\sigma_M = \sigma_P = 1.43$. Its wealthy EIS of 1.20 uses the wealth-weighted median in Calvet et al. (2026), while the borrower EIS is approximately their household-weighted median of 0.70. Their finding that EIS increases with initial wealth relative to income motivates the ordering; assigning these distributional summaries to our groups remains a calibration choice.[5]

Each alternative follows the main calibration procedure, including $\Sigma_R = 0.699\sigma_R$, the same initial wealth, interest-rate and debt targets, and the wealthy Euler equation that determines the fund holding cost. Prices and asset holdings at the reference are unchanged. Higher borrower curvature also leaves initial consumption and holding costs unchanged. Lower wealthy curvature instead reduces ϕ_{MF} from 0.146 to 0.031 and raises initial wealthy consumption; the flexible exposure costs are recovered from the same matching conditions. The observed historical series and information assumptions are unchanged.

Figure 8 and Table 1 show that reversing the EIS ordering preserves the constrained economy's historical fit advantage: the reduction in the average absolute top-share gap is 33%, compared with 34% in the baseline. Absolute errors rise modestly in both economies. The historical result therefore does not require wealthy households to be less willing to substitute consumption over time. Lowering wealthy consumption curvature under the same calibration procedure also makes asset accumulation cheaper through the lower holding cost, and both economies substantially overpredict wealth concentration. In this alternative, the constrained economy no longer fits wealth shares more closely than the flexible benchmark.

The constrained economy's delayed top-share peak and larger cumulative response to equity demand survive, but the cumulative advantage falls from 142% to 13% with higher borrower curvature and to 26% with lower wealthy curvature. Broad fund demand still generates closer cumulative responses across economies. Portfolio allocation frictions

[4]The baseline wealthy curvature is the simple average of Straub's three reported age-group curvatures, rather than an estimate for a particular wealth group. The consumption curvature of 3.7 in De Nardi et al. (2025) is estimated for retirees across income groups.

[5]The median values are reported in Table 4 and Internet Appendix Table IA.16 of the May 2025 author version.

Table 1: Historical fit and demand responses under alternative consumption curvatures

A. Historical wealth shares: perfect foresight				
		Mean absolute gap (pp)		Fit improvement
Calibration	Wealth group	Constrained	Flexible	(%)
Baseline	Top 10%	2.07	3.12	33.7
	Bottom 50%	1.22	1.95	37.2
Higher borrower curvature	Top 10%	2.25	3.34	32.6
	Bottom 50%	1.30	2.14	39.1
Higher wealthy EIS	Top 10%	3.80	3.49	-8.9
	Bottom 50%	1.80	1.71	-5.1

B. Cumulative top-share response: constrained/flexible		
Calibration	Equity demand	Fund demand
Baseline	2.42	1.05
Higher borrower curvature	1.13	1.09
Higher wealthy EIS	1.26	1.05

Note: Panel A uses the mean absolute gap between model and data wealth share changes over the full perfect-foresight sample. Fit improvement is $100(1 - \text{MAE}^{\text{con}}/\text{MAE}^{\text{flex}})$; negative values indicate worse constrained fit. Panel B divides the constrained cumulative top-share response by its flexible counterpart, summing the full linear impulse response. Each calibration applies the same marginal demand incentive in both economies and maintains the scaling rule $k = \phi_{MF}$. These equilibrium response ratios differ from the conditional inequality multiplier and from responses to identical equity price paths. Baseline curvatures are $(\sigma_R, \sigma_M, \sigma_P) = (4.1, 1.43, 1.43)$; the alternatives are $(4.1, 5, 5)$ and $(1/1.2, 1.43, 1.43)$.

continue to generate distinct borrowing and valuation dynamics, but their quantitative contribution depends on the calibration of households' saving behavior.

Under unanticipated historical shocks, higher borrower curvature worsens the wealth-share fit and produces a negative middle-class collateral multiplier in the constrained economy in 23 months, beginning in August 1993. These paths are therefore diagnostics of the maintained binding system; interpreting those episodes as equilibria requires allowing the constraint to become slack.

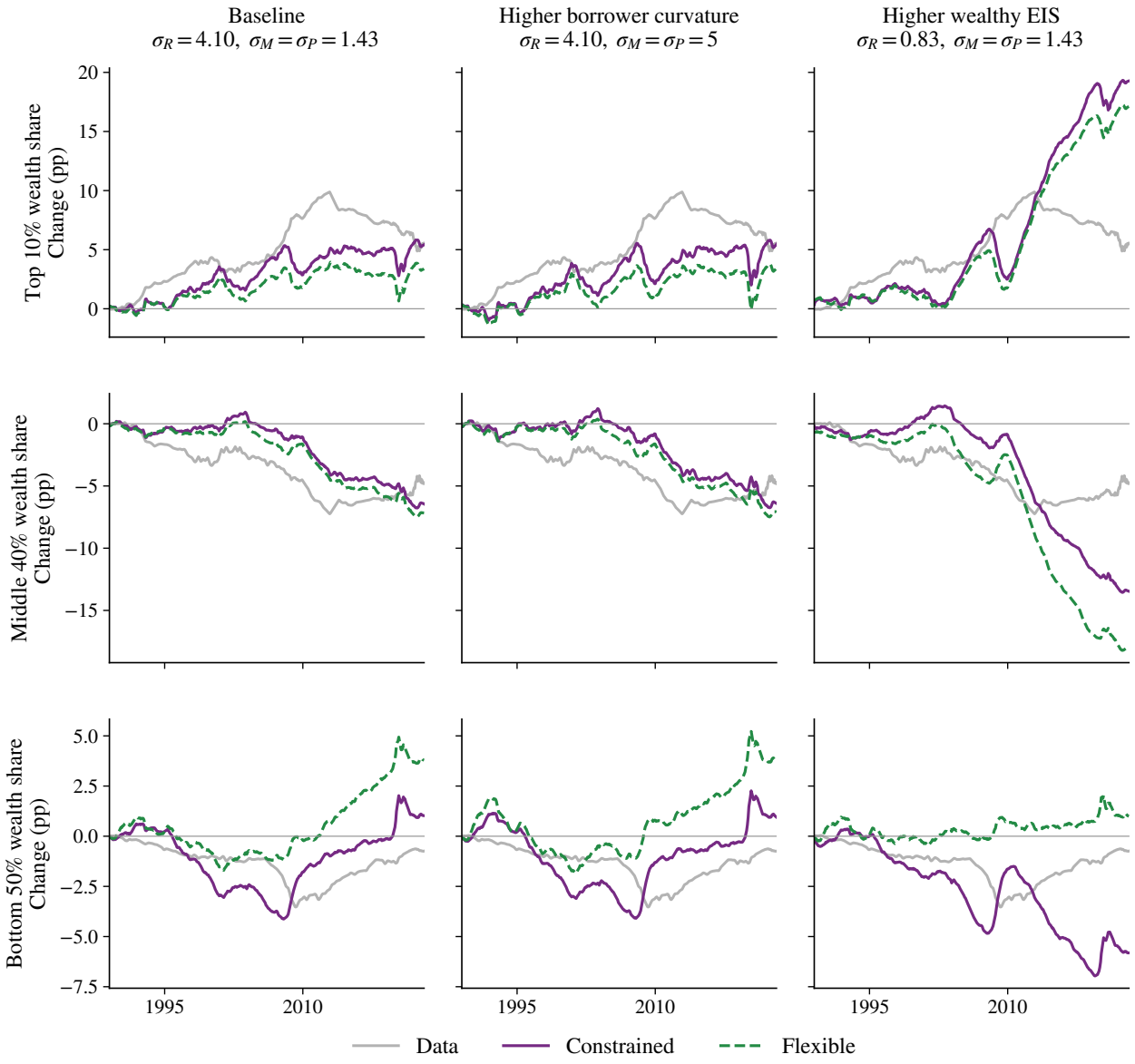


Figure 8: Historical wealth shares under alternative consumption curvatures

Note: Perfect-foresight simulations over January 1989–March 2023. Wealth share changes are in percentage points from the common 1989 wealth allocation. Gray: data; purple: constrained economy; green: flexible benchmark. Columns use the baseline, higher borrower curvature and higher wealthy EIS. Each follows the original calibration procedure with $\Sigma_R = 0.699\sigma_R$. Wealthy consumption and holding costs are recalculated in the third column. The baseline flexible path has three months in 2020 with a negative linear approximation to the middle-class collateral multiplier; both alternative perfect-foresight calibrations pass this check.

S4 Quantitative Model: Full Impulse Response Set

This section reports responses to the seven fundamental disturbances and the two demand disturbances of Section **??**. The single-innovation demand experiments shown here use the same marginal incentives in the constrained economy and the flexible benchmark, as specified in Appendix **??**. Each demand innovation is 0.01, with persistence 0.9; fundamental disturbances retain their calibrated monthly innovation scales. Solid red denotes the constrained economy and dashed blue the flexible benchmark. Month zero is the innovation month. The grids display 60 monthly observations. The flexible fund retains the exposure costs described in Appendix **??**. The selected dividend and borrowing responses appear first, followed by the complete grids.

S4.1 Dividend Disturbance

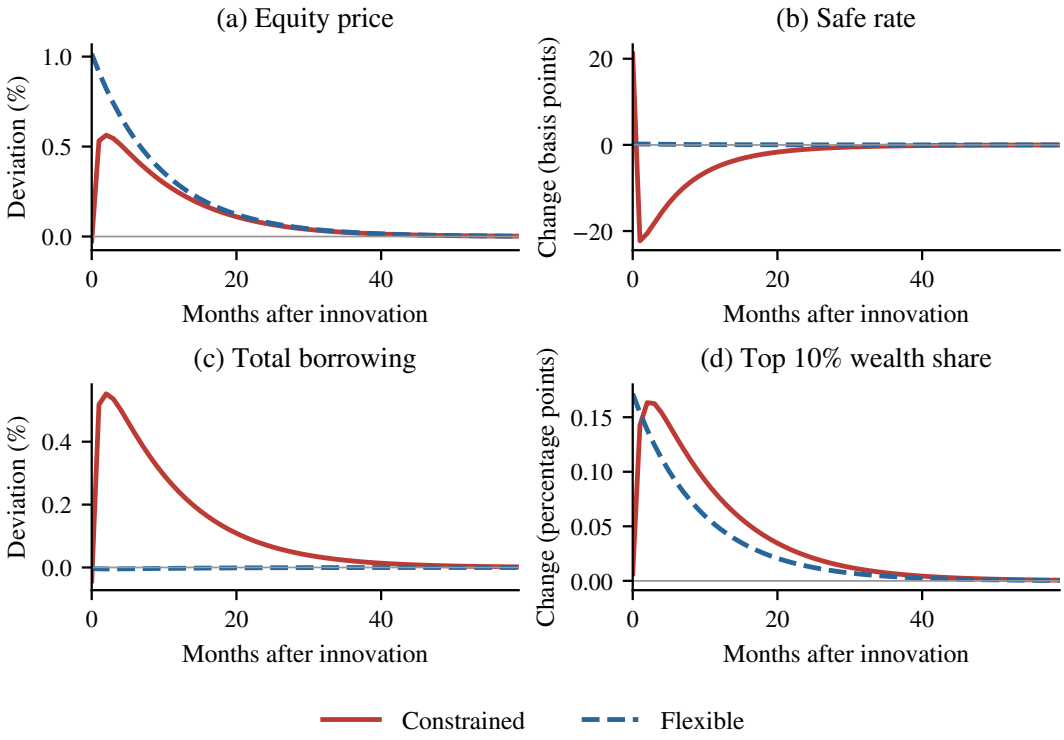


Figure 9: Dividend shock: headline responses

Note: Response to a positive one-standard-deviation dividend innovation in the baseline monthly specification. Solid red: constrained economy; dashed blue: flexible benchmark. Panel (a) equity price, in percent; (b) safe rate, in basis points; (c) total borrowing including government debt, in percent; (d) top 10% wealth share, in percentage points. Month zero is the innovation month.

S4.2 Borrowing Disturbances

Figures 10 and 11 show how public borrowing and a relaxation of middle-class collateral limits affect equity valuations. Both increase the fund's lending position. Maintaining its equity share then requires higher equity value in the constrained economy. In the flexible benchmark, the fund can change its allocation, and higher safe returns instead lower the equity price. This contrast illustrates why portfolio restrictions matter even when the disturbance originates in borrowing: credit conditions change wealth through both liabilities and the valuation of assets held at the top. The complete responses across shocks are reported below.

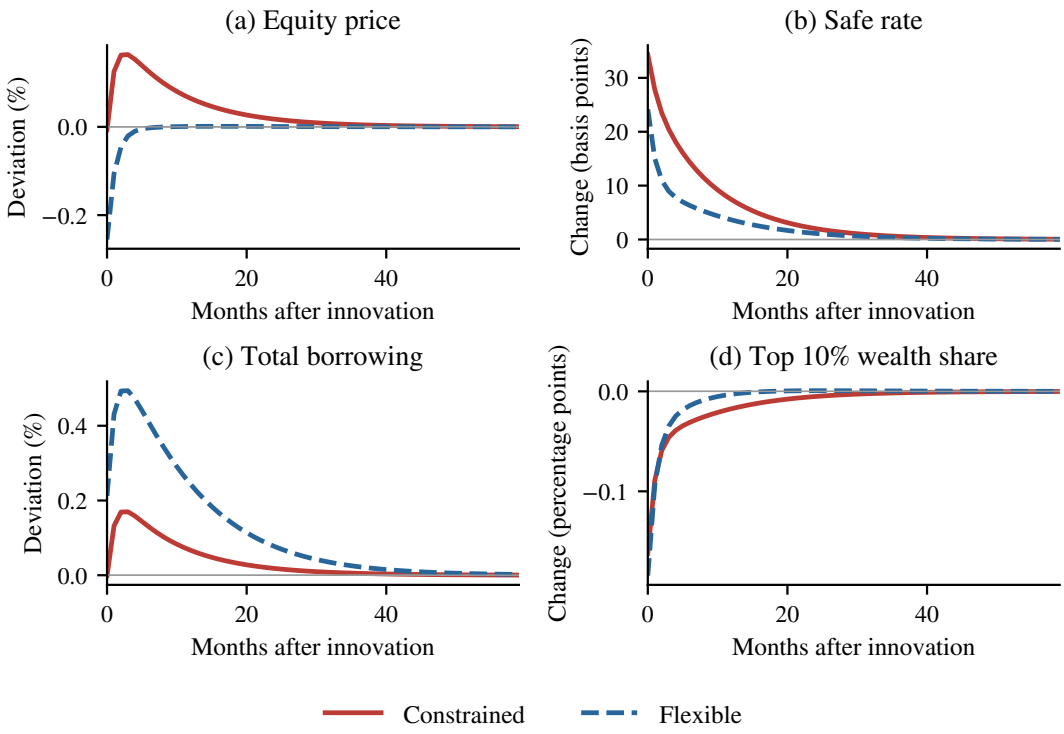


Figure 10: Impulse responses to a government-borrowing shock

Response to a one-standard-deviation innovation to government borrowing (B_t^G), constrained (solid red) versus flexible (dashed blue), over 60 months. Panel (a) equity price; (b) interest rate; (c) total borrowing; (d) top 10% wealth share. Price and total borrowing are percentage deviations from the reference; the safe rate changes in basis points and the wealth share in percentage points. Total borrowing includes government debt. Month zero is the innovation month.

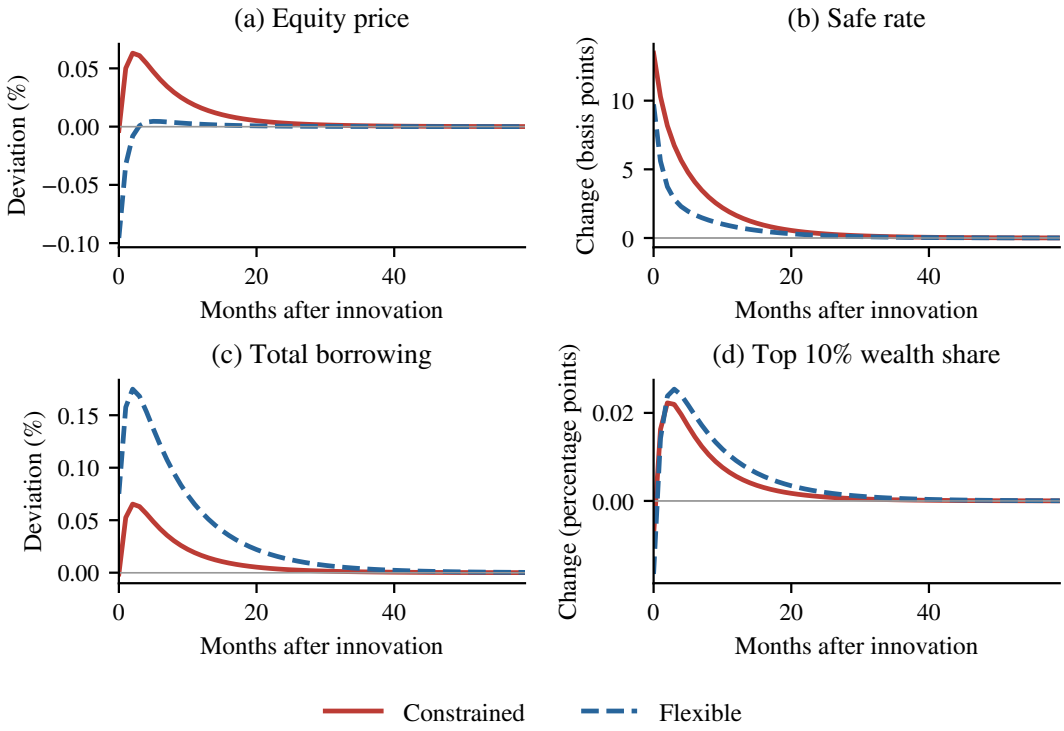


Figure 11: Impulse responses to a middle-class loan-to-value shock

Response to a one-standard-deviation innovation to the middle-class loan-to-value ratio (τ_t^M), constrained (solid red) versus flexible (dashed blue), over 60 months. Panels as in Figure 10.

S4.3 Complete Response Grids

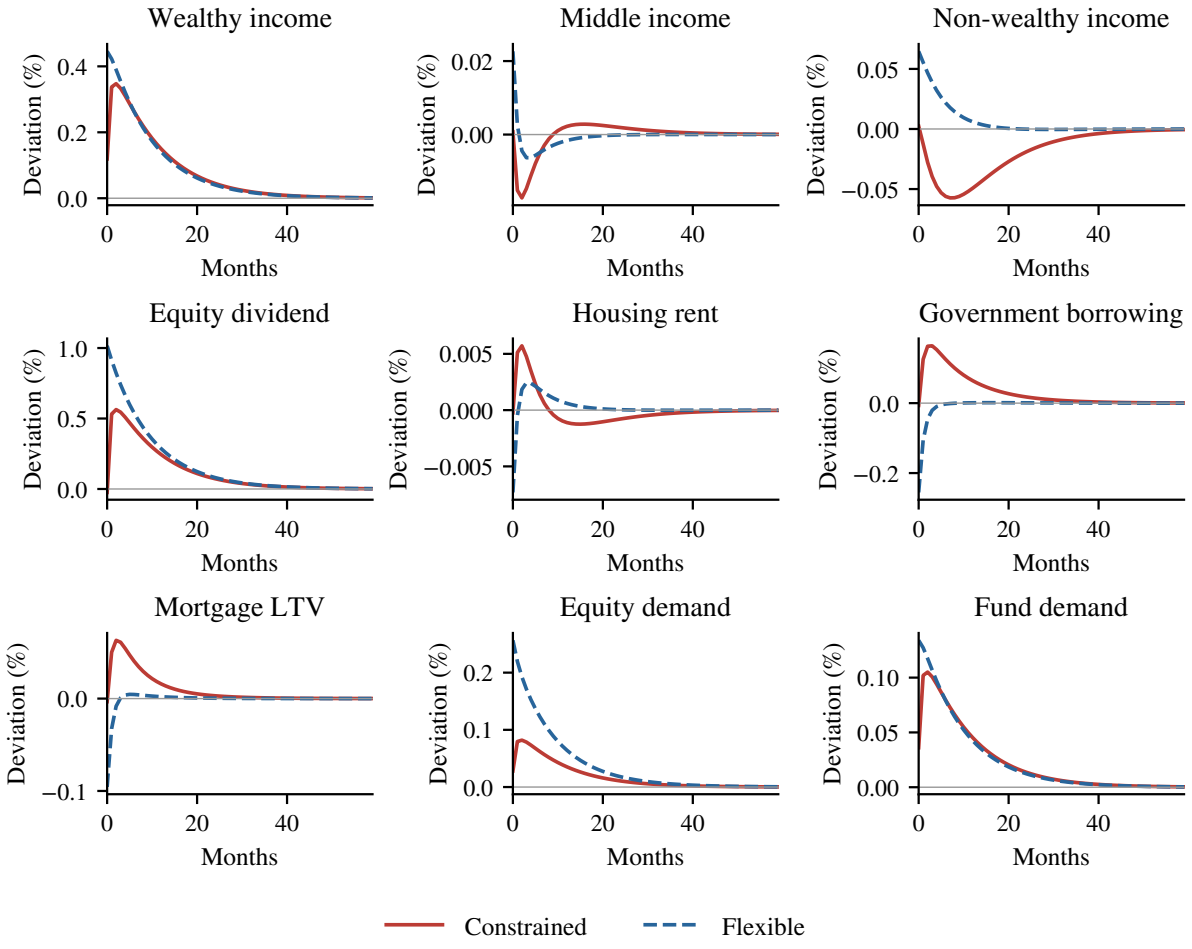


Figure 12: Equity price across all shocks

Note: Percentage-deviation response of the equity price to each shock, constrained (solid) versus flexible (dashed), over 60 months.

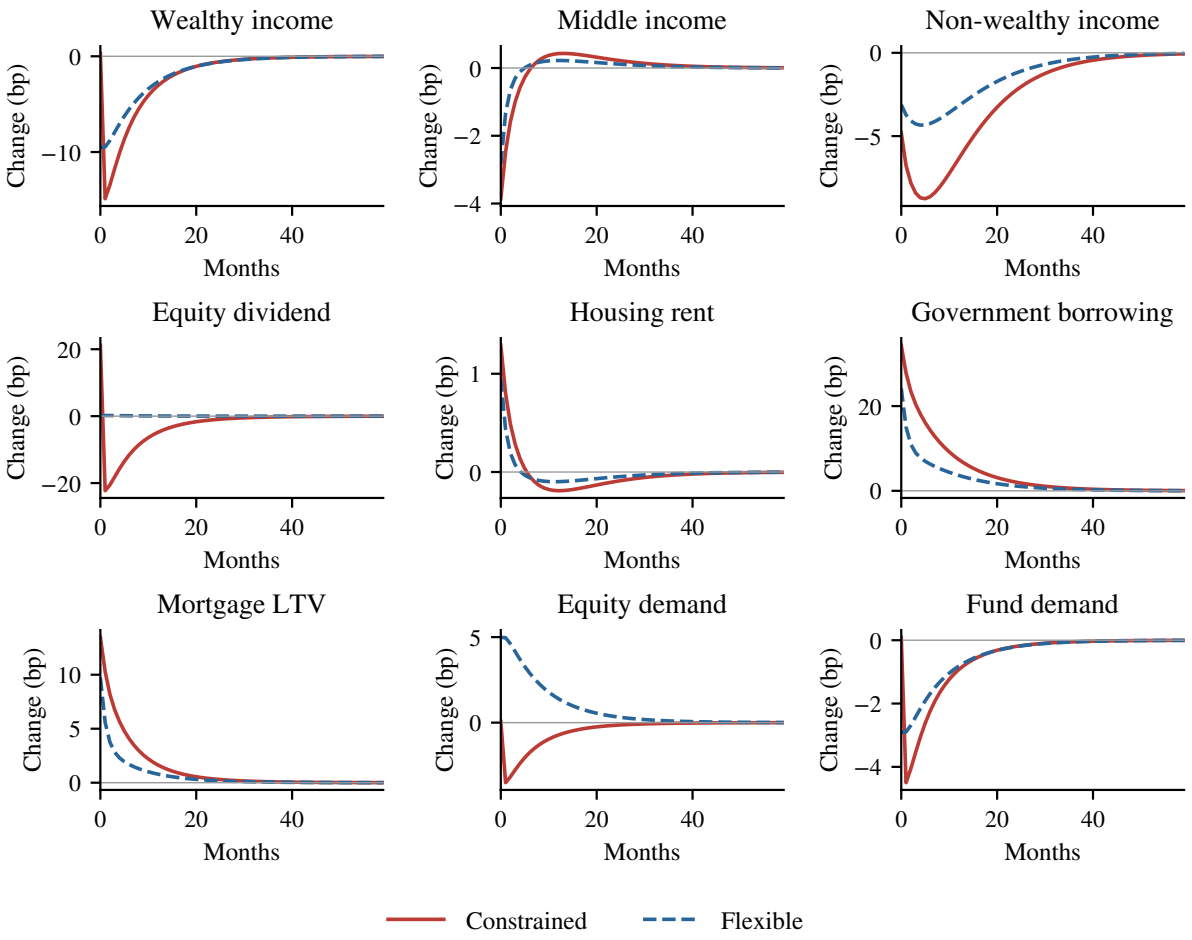


Figure 13: Safe interest rate across all shocks

Note: Change in the safe rate in basis points, $10^4 \bar{R}\widehat{R}_t$, to each shock, constrained (solid) versus flexible (dashed).

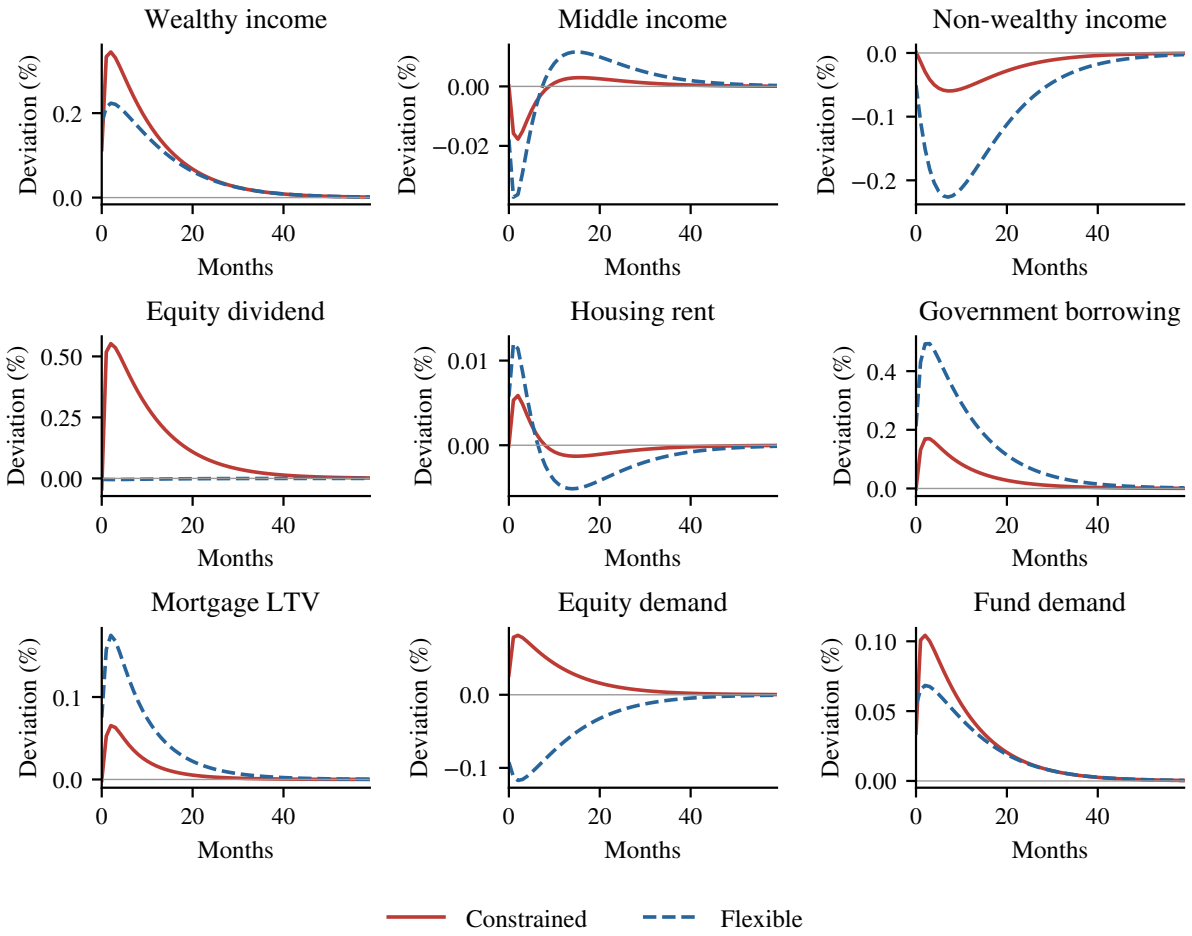


Figure 14: Total borrowing across all shocks

Note: Percentage-deviation response of total borrowing, including government debt, to each shock, constrained (solid) versus flexible (dashed).

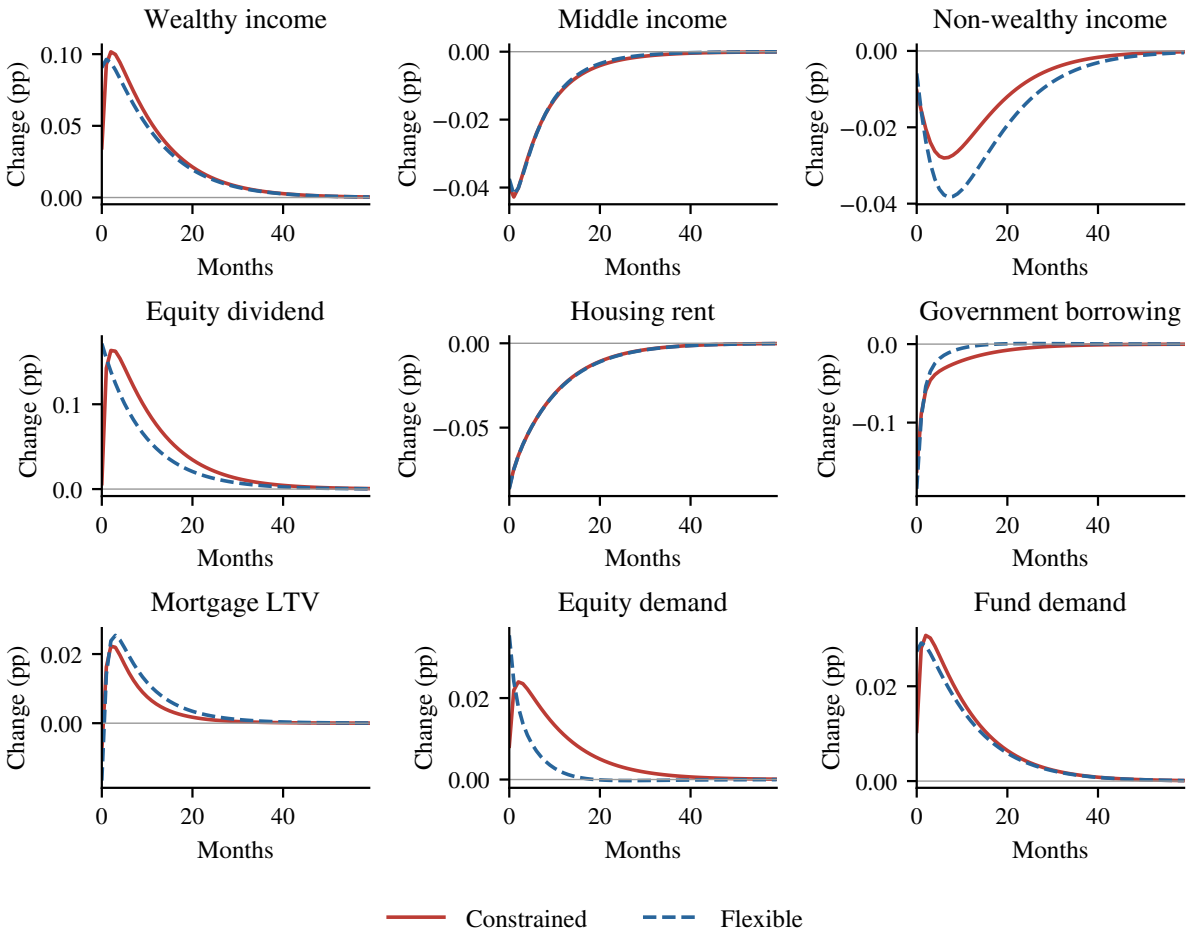


Figure 15: Top 10% wealth share across all shocks

Note: Response of the top 10% wealth share (percentage points) to each shock, constrained (solid) versus flexible (dashed).

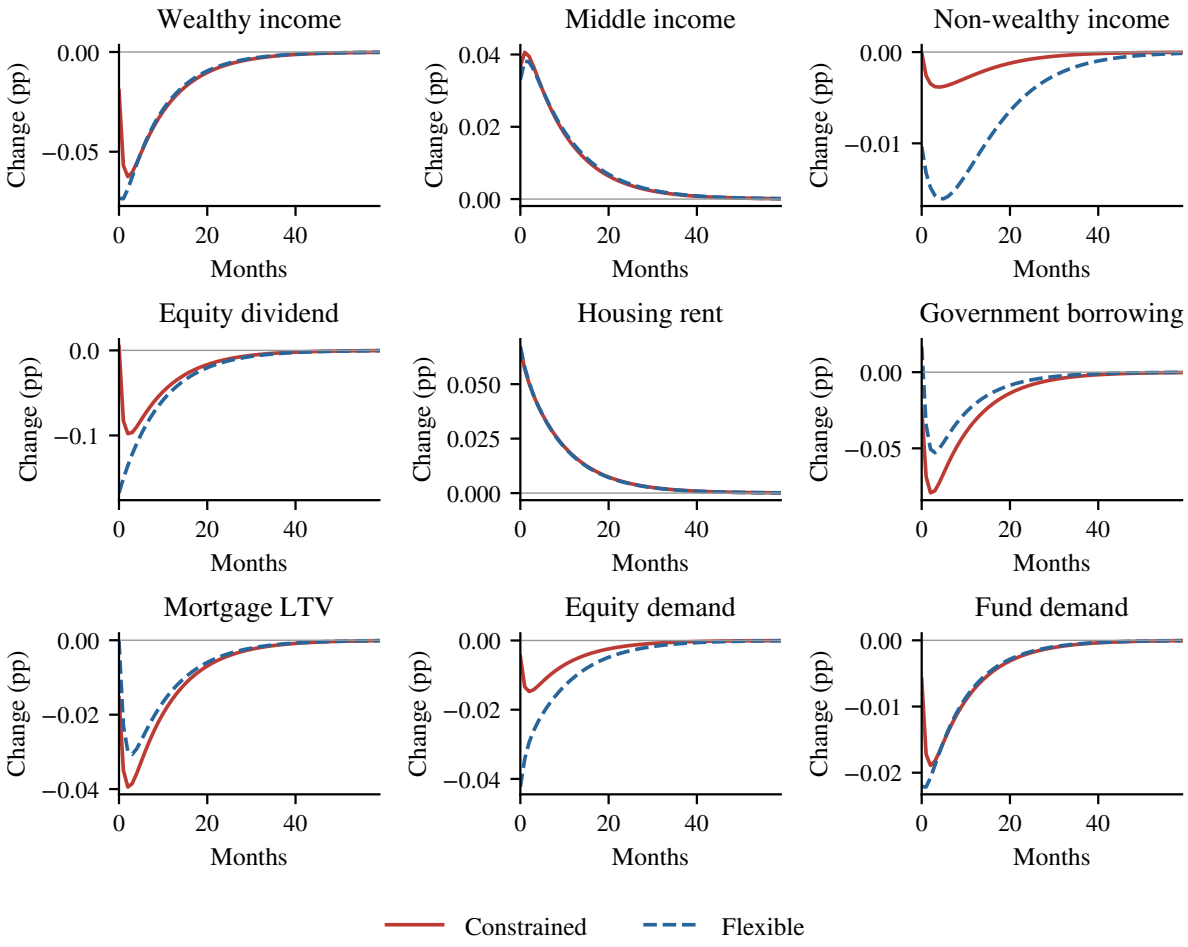


Figure 16: Middle 40% wealth share across all shocks

Note: Response of the middle 40% wealth share (percentage points) to each shock, constrained (solid) versus flexible (dashed).

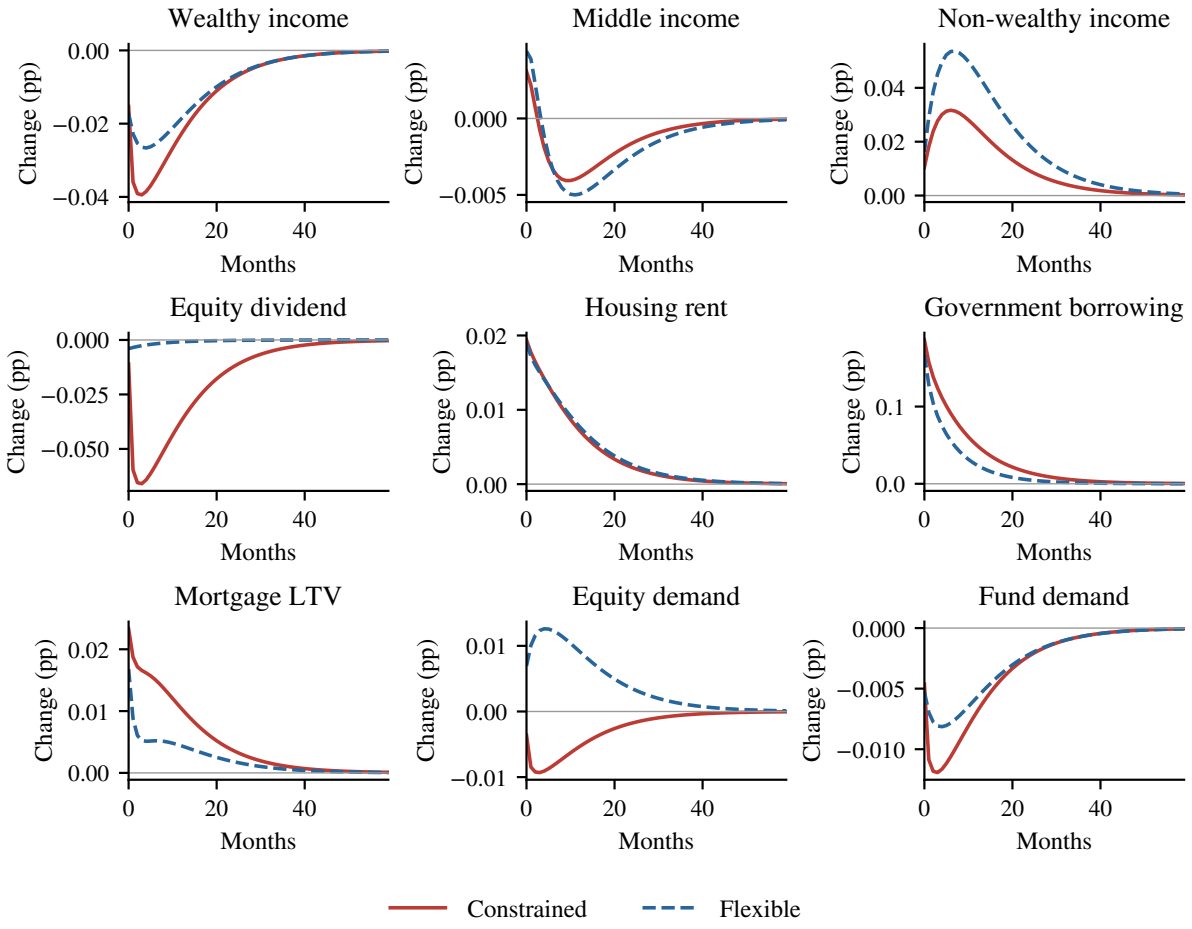


Figure 17: Bottom 50% wealth share across all shocks

Note: Response of the bottom 50% wealth share (percentage points) to each shock, constrained (solid) versus flexible (dashed).

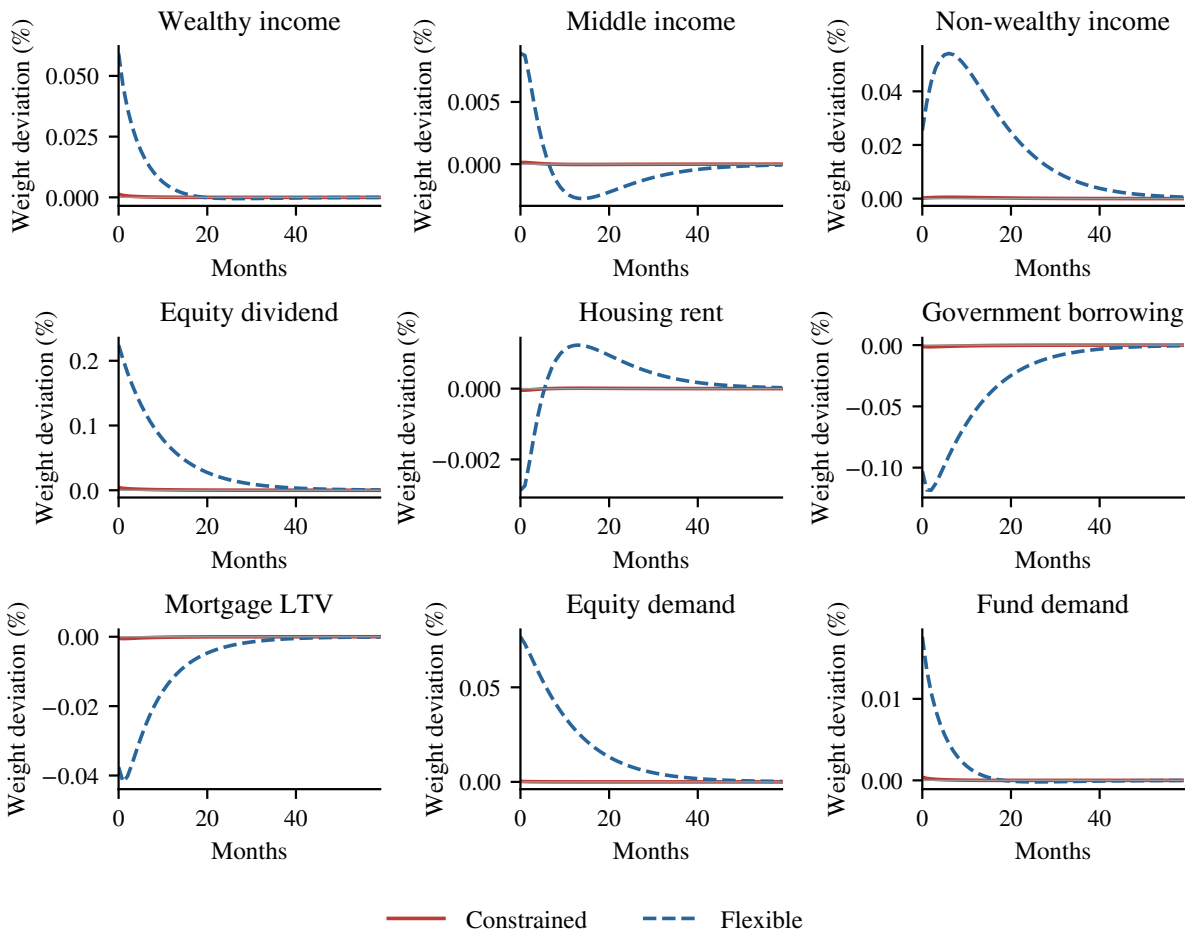


Figure 18: Wealthy portfolio equity share across all shocks

Note: Percentage deviation of the equity portfolio weight from its reference, constrained (solid) versus flexible (dashed). The plotted objects are $\widehat{\theta}$ for the constrained fund and $\widehat{\theta}^{\text{flex}}$ for the flexible fund, with $\widehat{\theta}^{\text{flex}} = \widehat{p} - \widehat{a}^{R}$. Equity quantity is fixed in these simulations.

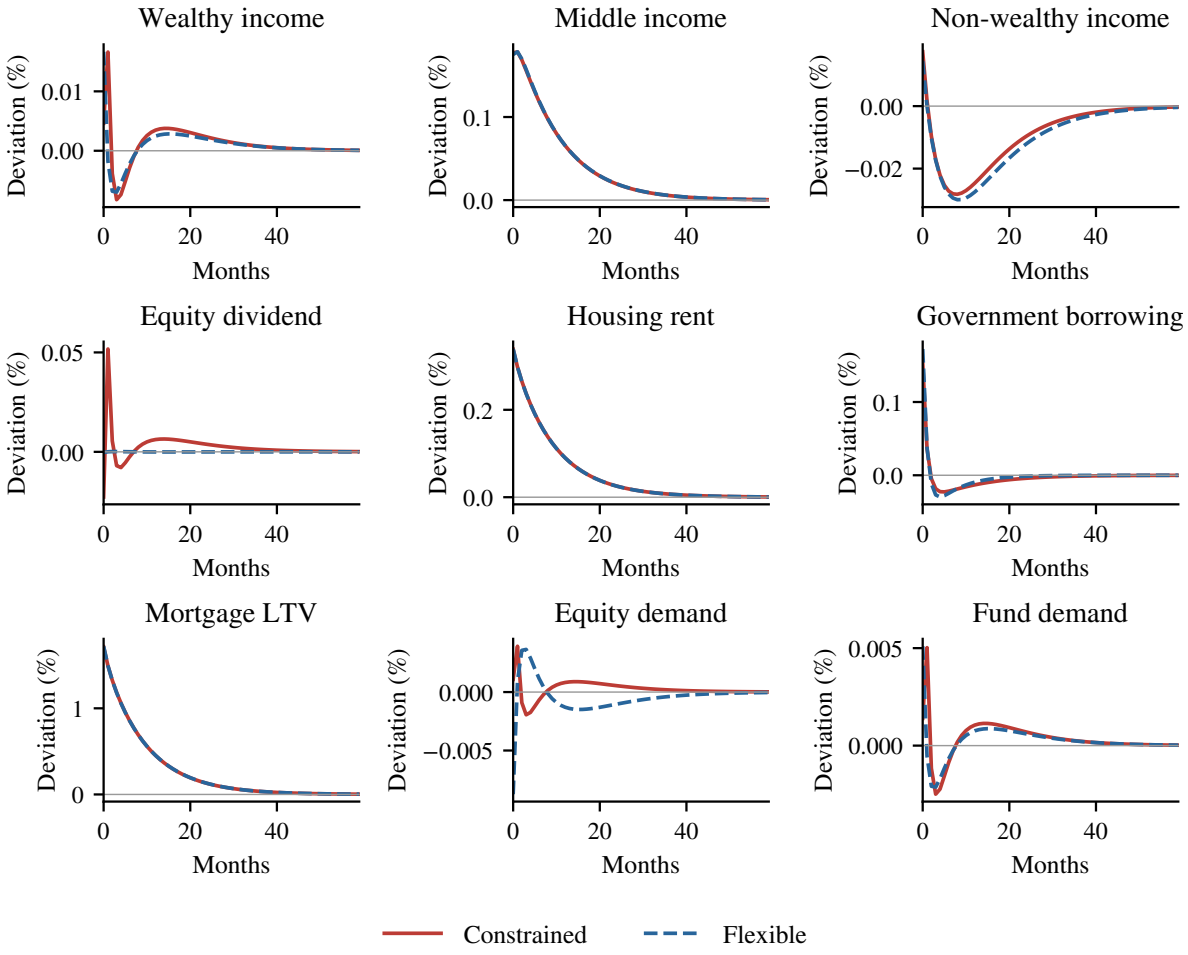


Figure 19: Middle-class borrowing across all shocks

Note: Percentage-deviation response of middle-class borrowing to each shock, constrained (solid) versus flexible (dashed).

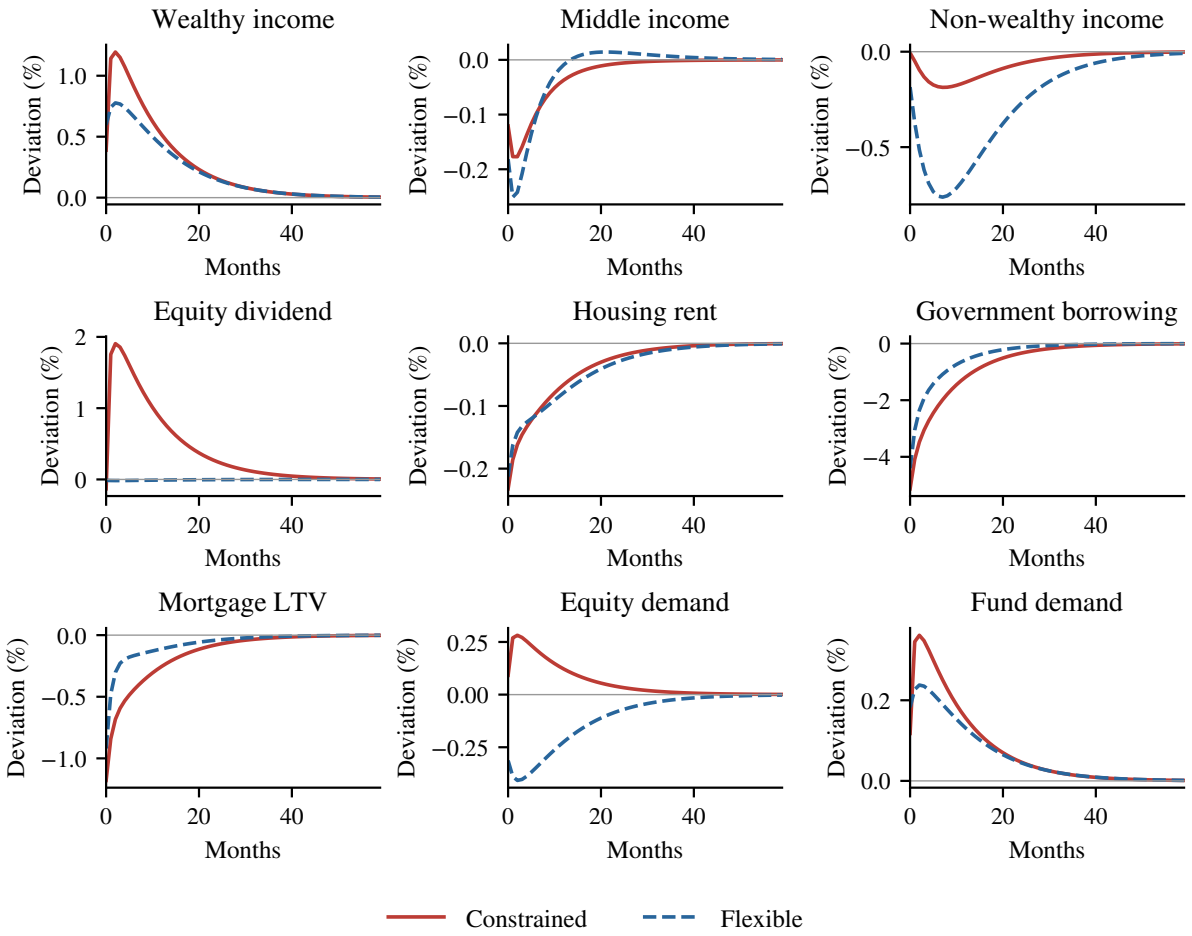


Figure 20: Non-wealthy borrowing across all shocks

Note: Percentage-deviation response of non-wealthy borrowing to each shock, constrained (solid) versus flexible (dashed).

References

Adrian, T. and H. S. Shin (2010): "Liquidity and leverage," *Journal of Financial Intermediation*, 19, 418–437.

Basak, S. and A. Pavlova (2013): "Asset prices and institutional investors," *American Economic Review*, 103, 1728–1758.

Brunnermeier, M. K. and Y. Sannikov (2014): "A macroeconomic model with a financial sector," *American Economic Review*, 104, 379–421.

Calvet, L. E., J. Y. Campbell, F. Gomes, and P. Sodini (2026): "The Cross-Section of Household Preferences," *The Journal of Finance*, 81, 2773–2808.

Chen, H., V. Cúrdia, and A. Ferrero (2012): "The macroeconomic effects of large-scale asset purchase programmes," *The Economic Journal*, 122, F289–F315.

De Nardi, M., E. French, J. B. Jones, and R. McGee (2025): "Why Do Couples and Singles Save during Retirement? Household Heterogeneity and Its Aggregate Implications," *Journal of Political Economy*, 133, 750–792.

Du, W., B. Hébert, and A. W. Huber (2023): "Are intermediary constraints priced?" *The Review of Financial Studies*, 36, 1464–1507.

Duffie, D. (2010): "Presidential address: Asset price dynamics with slow-moving capital," *The Journal of Finance*, 65, 1237–1267.

Gabaix, X. and R. S. J. Koijen (2021): "In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis," *Working paper*.

Gârleanu, N. and L. H. Pedersen (2018): "Efficiently inefficient markets for assets and asset management," *The Journal of Finance*, 73, 1663–1712.

Haddad, V., P. Huebner, and E. Loualiche (2025): "How Competitive Is the Stock Market? Theory, Evidence from Portfolios, and Implications for the Rise of Passive Investing," *American Economic Review*, 115, 975–1018.

Haddad, V. and T. Muir (2021): "Do intermediaries matter for aggregate asset prices?" *The Journal of Finance*, 76, 2719–2761.

He, Z. and A. Krishnamurthy (2013): "Intermediary asset pricing," *American Economic Review*, 103, 732–770.

——— (2018): "Intermediary asset pricing and the financial crisis," *Annual Review of Financial Economics*, 10, 173–197.

He, Z. and W. Xiong (2013): "Delegated asset management, investment mandates, and capital immobility," *Journal of Financial Economics*, 107, 239–258.

Kacperczyk, M., S. Van Nieuwerburgh, and L. Veldkamp (2016): "A rational theory of mutual funds' attention allocation," *Econometrica*, 84, 571–626.

Kaplan, G., B. Moll, and G. L. Violante (2018): "Monetary policy according to HANK," *American Economic Review*, 108, 697–743.

Pavlova, A. and T. Sikorskaya (2023): "Benchmarking intensity," *The Review of Financial Studies*, 36, 859–903.

Vayanos, D. and J.-L. Vila (2021): "A preferred-habitat model of the term structure of interest rates," *Econometrica*, 89, 77–112.